Code-Layout, Readability and Reusability

Date	9 July 2026
Team ID	Not Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained in Python & frontend
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for frontend, backend, and ML model
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Used names like soil_data, prediction_result
4	Function / Method Names	Functions are descriptively named	Yes	Functions like predict_crop(), process_input()
5	Code Comments	Inline and block comments explain logic	Partial	Basic comments added, can improve more
6	Modular Design	Code is split into reusable functions /modules	Yes	Separate modules for ML model and API
7	No Redundant Code	Duplicate or unused code is removed	Yes	Clean and optimized code structure
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation done, advanced handling can be added

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Crop Prediction Function (predict_crop)	Python (Scikit-learn)	Backend API	High

2	Input Processing Module	Python	Before prediction	High
3	Frontend Form Component	HTML/CSS/JS	User input page	Medium
4	API Endpoint (/predict)	Flask	Communication between frontend & backend	High
5	Dataset Loader	Python (Pandas)	Model training & testing	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Well-organized with clear separation of components
Readability	4	Easy to understand with meaningful names
Reusability	4	Modular functions can be reused in future
Documentation / Comments	4	Basic comments present, can be improved
Overall Score	4	Good quality code with scope for minor improvements